

Where Do Museum Visitors Go?

Modeling Foot Traffic with Markov Chains

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The setting

The National Museum of History in Taipei is a Japanese-style building established in 1955. On the day we collected data, only the third floor was open to visitors. We used that floor as the basis for a Markov chain model to predict how visitors move between rooms over time.

Mapping the floor

The third floor is divided into three sections (301, 302, 303), each split into a front and back area, giving six rooms total. We represent these as nodes in a graph, with edges connecting rooms that share a direct path (Figure 1). Movement is only possible between connected rooms: for example, you cannot walk directly from room 1 to room 3. Room 5 has the most connections, linking to rooms 3, 4, and 6.



Figure 1: Map of the third floor of the National Museum of History, Taipei. The floor is divided into three sections: 301, 302, and 303.

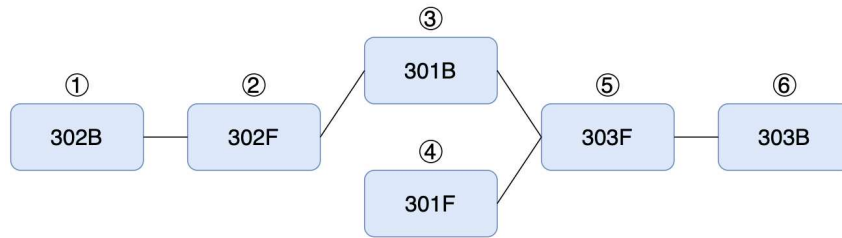


Figure 2: Each section is divided into front (F) and back (B) subsections, labeled as rooms 1 through 6.

Data collection

At the start of each 5-minute interval, we counted the number of people in each room and tracked where they moved by the end of the interval. Since the floor was not crowded (1 to 29 people per room at any time), we were able to accurately record both how many people left each room and which room they went to.

We recorded the results in matrix A , where each column represents a starting room and each row represents a destination. Entry A_{ij} is the number of people who moved from

room j to room i during the interval.

$$A = \begin{bmatrix} 5 & 2 & 0 & 0 & 0 & 0 \\ 2 & 6 & 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 10 & 0 \\ 0 & 0 & 3 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 13 & 1 \end{bmatrix}$$

We made two assumptions to keep the model tractable: (1) the third floor operates as a closed system during the observation window, meaning no one enters or exits the floor, and (2) within any 5-minute interval, a person either stays in their room or moves to an adjacent one.

Building the Markov matrix

To convert A into a transition matrix M , we divide each entry by the total number of people in that column's starting room at the beginning of the interval. Each column of M then sums to 1, making it a valid stochastic matrix.

$$M = \begin{bmatrix} \frac{5}{7} & \frac{1}{6} & 0 & 0 & 0 & 0 \\ \frac{2}{7} & \frac{1}{2} & \frac{2}{5} & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 & \frac{10}{29} & 0 \\ 0 & 0 & \frac{3}{5} & 0 & \frac{2}{29} & 0 \\ 0 & 0 & 0 & 1 & \frac{4}{29} & \frac{2}{3} \\ 0 & 0 & 0 & 0 & \frac{13}{29} & \frac{1}{3} \end{bmatrix}$$

Does the system reach a steady state?

M has six eigenvalues. One of them is $\lambda = 1$, with associated eigenvector

$$v = (1, 12/7, 10/7, 36/35, 87/35, 117/70)^T$$

The remaining eigenvalues are either between 0 and 1, or complex with modulus less than 1. Normalizing v so its entries sum to 1 gives the steady-state distribution:

$$v_{\text{eigen}} = \begin{pmatrix} 0.107 \\ 0.184 \\ 0.153 \\ 0.110 \\ 0.266 \\ 0.179 \end{pmatrix}$$

We verify this using three approaches.

Empirical approach

We test three different starting distributions of visitors across the six rooms:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1/3 \\ 0 \\ 1/3 \\ 0 \\ 1/3 \\ 0 \end{pmatrix}$$

Multiplying each by M^t for increasing t shows how the distributions evolve. After $t = 12$ intervals (1 hour), the three distributions are still noticeably different:

$$M^{12}v_1 = \begin{pmatrix} 0.174 \\ 0.236 \\ 0.150 \\ 0.103 \\ 0.208 \\ 0.128 \end{pmatrix}, \quad M^{12}v_2 = \begin{pmatrix} 0.112 \\ 0.187 \\ 0.153 \\ 0.110 \\ 0.263 \\ 0.176 \end{pmatrix}, \quad M^{12}v_3 = \begin{pmatrix} 0.122 \\ 0.196 \\ 0.152 \\ 0.109 \\ 0.253 \\ 0.168 \end{pmatrix}$$

By $t = 36$ (3 hours), all three have converged to v_{eigen} :

$$M^{36}v_1 = \begin{pmatrix} 0.108 \\ 0.185 \\ 0.153 \\ 0.110 \\ 0.265 \\ 0.178 \end{pmatrix}, \quad M^{36}v_2 = \begin{pmatrix} 0.107 \\ 0.184 \\ 0.153 \\ 0.110 \\ 0.266 \\ 0.179 \end{pmatrix}, \quad M^{36}v_3 = \begin{pmatrix} 0.107 \\ 0.184 \\ 0.153 \\ 0.110 \\ 0.266 \\ 0.179 \end{pmatrix}$$

The intuition behind this: any starting vector v_i has a component in the direction of v_{eigen} . Repeated multiplication by M leaves that component unchanged (since $\lambda = 1$), while all other components shrink toward zero because their eigenvalues have modulus less than 1.

Regularity approach

Raising M to a high power directly confirms convergence. By M^{40} , every column has become identical:

$$M^{40} = \begin{bmatrix} 0.108 & 0.107 & 0.107 & 0.107 & 0.107 & 0.107 \\ 0.184 & 0.184 & 0.184 & 0.184 & 0.184 & 0.184 \\ 0.153 & 0.153 & 0.153 & 0.153 & 0.153 & 0.153 \\ 0.110 & 0.110 & 0.110 & 0.110 & 0.110 & 0.110 \\ 0.266 & 0.266 & 0.266 & 0.266 & 0.267 & 0.267 \\ 0.179 & 0.179 & 0.179 & 0.179 & 0.179 & 0.179 \end{bmatrix}$$

Since any starting distribution sums to 1, multiplying M^{40} by any v_i yields the same result: v_{eigen} .

Graph approach

The graph in Figure 2 is aperiodic (no fixed cycle forces the system to revisit rooms at regular intervals) and strongly connected (there is a directed path between every pair of rooms). Together, these two properties guarantee that the Markov chain converges to a unique steady state regardless of where visitors start.

Summary

The system reaches a steady-state distribution after roughly 3 hours (36 five-minute intervals). In the long run, visitors spread across the six rooms as follows:

$$v_{\text{dist}} = \begin{pmatrix} 0.11 \\ 0.18 \\ 0.15 \\ 0.11 \\ 0.27 \\ 0.18 \end{pmatrix}$$

Room 5 attracts the most visitors (27%), likely because it connects to the most other rooms. Rooms 1 and 4 each hold around 11%, consistent with their position at the ends of the network.

Appendix

Appendix A: Matrices A and M

```
A = matrix([[5, 2, 0, 0, 0, 0],
            [2, 6, 2, 0, 0, 0],
            [0, 4, 0, 0, 10, 0],
            [0, 0, 3, 0, 2, 0],
            [0, 0, 0, 1, 4, 2],
            [0, 0, 0, 0, 13, 1]])

M = matrix([[5/7, 2/12, 0, 0, 0, 0],
            [2/7, 6/12, 2/5, 0, 0, 0],
            [0, 4/12, 0, 0, 10/29, 0],
            [0, 0, 3/5, 0, 2/29, 0],
            [0, 0, 0, 1, 4/29, 2/3],
            [0, 0, 0, 0, 13/29, 1/3]])

A, M
```

```
(
[ 5  2  0  0  0  0] [ 5/7  1/6  0  0  0  0]
[ 2  6  2  0  0  0] [ 2/7  1/2  2/5  0  0  0]
[ 0  4  0  0 10  0] [ 0  1/3  0  0 10/29  0]
[ 0  0  3  0  2  0] [ 0  0  3/5  0  2/29  0]
[ 0  0  0  1  4  2] [ 0  0  0  1  4/29  2/3]
[ 0  0  0  0 13  1], [ 0  0  0  0 13/29  1/3]
)
```

Appendix B: Eigenvalues and eigenvectors of M

```
M.eigenvectors_right()
```

```
[(1,
 [(1, 12/7, 10/7, 36/35, 87/35, 117/70)],
 1),
 (0.1771873785016486?,
 [(1, -3.222590014704395?, 1.886446112367128?, 7.977757230534017?,
 4.084508227418958?, -11.72612155561571?)],
 1),
 (0.4813533671531864?,
 [(1, -1.397594082795167?, -0.6491346549588127?, -0.7453993448188359?,
 0.4448644728579472?, 1.347263609714869?)],
 1),
 (0.843727807065420?,
 [(1, 0.7766525566782319?, -0.04689301388881387?, -0.1040923774998686?,
 -0.8655024634700727?, -0.7601647018194769?)],
 1),
 (-0.4083592353092242? - 0.4003813587568928?*I,
 [(1, -6.735869697569631? - 2.402288152541357?*I, ...)],
 1),
 (-0.4083592353092242? + 0.4003813587568928?*I,
 [(1, -6.735869697569631? + 2.402288152541357?*I, ...)],
 1)]
```

Appendix C: Normalizing the steady-state eigenvector

```
eigen_sum = 1 + 12/7 + 10/7 + 36/35 + 87/35 + 117/70
v_eigen = vector([1/eigen_sum, 12/7/eigen_sum, 10/7/eigen_sum,
                  36/35/eigen_sum, 87/35/eigen_sum, 117/70/eigen_sum])
v_eigen.n(digits=3)
```

```
(0.107, 0.184, 0.153, 0.110, 0.266, 0.179)
```

Appendix D: Initial distributions and long-term convergence

```
v1 = vector([1, 0, 0, 0, 0, 0])
v2 = vector([1/6, 1/6, 1/6, 1/6, 1/6, 1/6])
v3 = vector([1/3, 0, 1/3, 0, 1/3, 0])

# t=2 (10 minutes)
M^2*v1.n(digits=3), M^2*v2.n(digits=3), M^2*v3.n(digits=3)
```

```
((0.558, 0.347, 0.0952, 0.000, 0.000, 0.000),
 (0.138, 0.186, 0.170, 0.0886, 0.240, 0.178),
 (0.208, 0.228, 0.0920, 0.0721, 0.329, 0.0704))
```

```
# t=12 (1 hour)
M^12*v1.n(digits=3), M^12*v2.n(digits=3), M^12*v3.n(digits=3)
```

```
((0.174, 0.236, 0.150, 0.103, 0.208, 0.128),
 (0.112, 0.187, 0.153, 0.110, 0.263, 0.176),
 (0.122, 0.196, 0.152, 0.109, 0.253, 0.168))
```

```
# t=36 (3 hours)
M^36*v1.n(digits=3), M^36*v2.n(digits=3), M^36*v3.n(digits=3)
```

```
((0.108, 0.185, 0.153, 0.110, 0.265, 0.178),
 (0.107, 0.184, 0.153, 0.110, 0.266, 0.179),
 (0.107, 0.184, 0.153, 0.110, 0.266, 0.179))
```

Appendix E: Regularity approach

```
(M^40).n(digits=3)
```

```
[0.108 0.107 0.107 0.107 0.107 0.107]
[0.184 0.184 0.184 0.184 0.184 0.184]
[0.153 0.153 0.153 0.153 0.153 0.153]
[0.110 0.110 0.110 0.110 0.110 0.110]
[0.266 0.266 0.266 0.267 0.267 0.267]
[0.179 0.179 0.179 0.179 0.179 0.179]
```
